

PHYSICS LABORATORY MANUAL

(B.Tech.)

BIT Mesra, Jaipur Campus

Birla Institute of Technology Mesra, Jaipur Campus
Department of Physics

List of Physics Laboratory Experiments
Session:- Spring-2019

Course code: PH 114

Course title: PHYSICS LAB

1. To determine the frequency of AC mains with the help of Sonometer
2. To determine the wavelength of sodium light by Newton's rings Method
3. To determine the resistance per unit length of a Carey Foster's bridge wire and then to find the resistivity of the material of a given wire.
4. Measurement of voltage and frequency of a given signal using cathode ray oscilloscope
5. To determine the wavelength of prominent spectral lines of mercury light by a plane transmission grating using normal incidence
6. To calibrate a voltmeter using a potentiometer
7. To study LR circuit with an AC and determine the inductance of an inductor, its internal resistance and power factor of circuit.
8. To study the single slit diffraction and measure the slit width.
9. To determine the wavelength of sodium light by Fresnel's Biprism
10. To determination the refractive index of the material of a prism using spectrometer and sodium light



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EXPERIMENT No. 01

Objective: To determine the frequency of AC Mains with the help of sonometer.

Apparatus: Sonometer with ~~non~~ ^{steel} magnetic wire (~~nichrome~~), Ammeter, step down transformer (2-10 Volts), Key, Horse shoe magnet, Wooden stand for mounting the magnet, Set of 50 gm masses, Screw gauge and meter scale (fitted with the sonometer)

Description of the apparatus: As shown in the given figure below, an uniform nichrome (non-magnetic) wire is stretched on a hollow wooden box (sonometer), one side of which is tied to the hook H, while the other passes over a frictionless pulley P. A hanger carrying masses is also attached to this end of the non-magnetic wire. A permanent strong horse shoe magnet NS is kept at the middle of the nichrome wire in such a way that it produces a magnetic field perpendicular to the direction of current, to be flown in the nichrome wire. Two moveable sharp edged bridges A and B are provided on the wooden box for stretching wire. A step down transformer (2-10V) is connected across the wire.

Working Principle: Let a sonometer wire stretched under a constant load be placed in an uniform magnetic field applied at the right angles to the sonometer wire in the horizontal plane and let an alternating current of low voltage (by means of the step down transformer) be passed through the wire. On account of interaction, between the magnetic field and the current in the wire ($F = il \times B$), the wire will be deflected. The direction of deflection is being given by the Fleming's left hand rule. As the current is alternating, for half the cycle the wire will move upwards and for the next half the wire will move downwards. Therefore the sonometer wire will receive impulses alternately in opposite directions at the frequency of the alternating current passing through the wire. As a consequence the wire will execute forced vibrations with a frequency of the AC mains (under the conditions of resonance) in the sonometer wire.

The frequency of AC Mains, which is equal to the frequency of vibration of the sonometer wire in its fundamental mode (only one loop between the two bridges A and B, i.e., having two nodes and one antinode between the two bridges) is given by (under resonance conditions):

$$f = n = \frac{1}{2l} \sqrt{\frac{T}{m}}$$

$$\begin{aligned} f &= \text{freq. of AC mains} \\ n &= \text{freq. of sonometer wire} \end{aligned} \quad (1)$$

where T is the tension applied on the wire and given by $T = Mg$, M being the total mass loaded on the wire (i.e., total mass kept on the hanger and the mass of the hanger) and g the acceleration due to gravity. Symbol l presents the length of the sonometer wire between the two bridges. The mass per unit length of the sonometer wire is represented by symbol m and can be calculated in terms of the radius r of the sonometer wire, and the density d of the material wire (~~nichrome~~) as

$$m = \pi r^2 \cdot d$$

Sonometer Experiment with Electromagnet

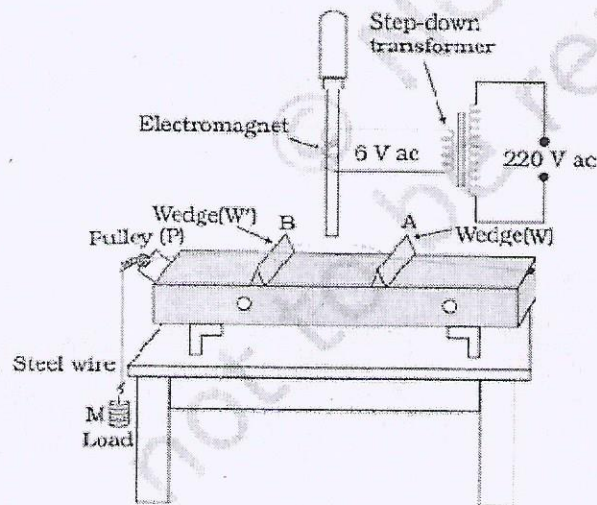


Fig 1: Experimental Set-up for measurement of AC frequency by Sonometer and Electromagnet

Sonometer Wire: Magnetic Wire(Steel)

Density of wire material (d) = 7.8 gram/CC

PRINCIPLE

The frequency n of the fundamental mode of vibration of a stretched string, fixed at two ends, is given by

$$n = \frac{1}{2l} \sqrt{\frac{T}{m}} \quad \text{--- (2)}$$

- If an alternating current is passed in the coil of the electromagnet, the magnetisation produced in the core is proportional to instantaneous value of the current. If the electromagnet is held close to the middle of the sonometer wire, the wire will be attracted twice during each cycle towards the electromagnet. The attractive force experienced by the wire will be proportional to the magnetisation produced in the core of the electromagnet. Since in each cycle, the wire will be pulled twice and hence at resonance, it will vibrate with a frequency which is twice the frequency of alternating current. Hence, if f is the frequency of the alternating current, then

$$f = \frac{n}{2} = \frac{1}{4l} \sqrt{\frac{T}{m}} \quad \text{--- (3)}$$

Freq. of AC mains (using Electromagnet) is given by,

$$f = \frac{1}{4l} \sqrt{\frac{T}{m}}$$

$$m = 3.14 r^2 d$$

(2)

Substitution of value of m , evaluated from the equation 2, in equation 1, gives the value of frequency of AC mains.

Methodology:

- Measure the diameter of the wire with a screw gauge at several points along its length. At each point two mutually perpendicular diameters should be measured. Evaluate the radius of the sonometer wire. [See observation table (a)]
- Connect the step down transformer to AC mains and connect the transformer output (6 Volts connection) to the two ends of the sonometer wire through a rheostat, ammeter and a key, as shown in the figure.
- Place the two movable sharp-edged bridges A and B at the two extremities of the wooden box.
- ~~X~~ Mount the horse shoe magnet vertically at the middle of the sonometer wire such that the wire passes freely in between the poles of the magnet and the face of the magnet is normal to the length of the wire. The direction of current flowing through the wire will now be normal to the magnetic field. (use electromagnet)
- Apply a suitable tension to the wire, say by putting 100 gm masses on the hanger [tension in the wire = (mass of the hanger + mass kept on the hanger) $\times g$]. Switch on the mains supply and close the key K and then adjust the two bridges A and B till the wire vibrates with the maximum amplitude (in the fundamental mode of resonance) between the two bridges. Measure the distance between the two bridges (l). [See observation table (b)]
- Increasing the load M by steps of 50 gm, note down the corresponding values of l for maximum amplitude (in the fundamental mode of resonance). Take six or seven such observations.
- Knowing all the parameters, using the relations given in equations 1 and 2 calculate the frequency of AC mains for each set of observation separately and then take mean
- Also plot a graph between the mass loaded, M along the X-axis and the square of the length l along Y-axis. This graph should be a straight line. Find the slope of this line and then using the equations 1 and 2, calculate the frequency of AC mains from this graph also. (Frequency = $\sqrt{g/(4 \times \text{slope} \times m)}$)

Observations:

[a] Measurement of radius of sonometer wire (r)

Least count of screw gauge = .. cm

Zero error of the screw gauge = .. cm

S.No.	Diameter of wire along one direction. cm.	Diameter of wire in Cm	Mean observed diameter cm.	mean corrected diameter cm.	mean radius r cm
1.					
2.					
3.					
4.					
5.					

[b] Measurement of T. / and frequency of the AC Mains:

Mass of the hanger = 50 gm

Acceleration due to gravity(g)= 980 cm/sec².

Density of sonometer wire (nichrome) = ~~8.18348 gm/cc~~

Steel = 7.8 gm/cc

Sl. No	Total Mass Loaded = Mass of hanger + Mass on it M gm.	Mass in wire T = M × g Gm-cm/s ²	Position of first bridge a cm.	Position of second bridge b cm	Length of wire between two bridges l=a-b cm	Frequency in Hz
1						
2						
3						
4						
5						
6						
7						

Mean value of the AC Mains frequency = .. Hz.

CALCULATIONS:

Calculations from the graph are also to be given on the left side of paper.

The slope of graph plotted between Mass loaded (M) and the square of length l

S = units

AC mains frequency when calculated from the graph = $\sqrt{[g/(4Sm)]}$

= Hz

Result:

The frequency of AC Mains as calculated:

- A. Experimental calculations: Hz.
 - B. Graphical calculations: Hz (Graph is attached)
 - C. Standard Value: .. 50 Hz (in this country)
- Percentage Error : %

Sources of errors and precautions:

- The sonometer wire should be uniform and without kinks.
- The pulley should be frictionless
- The wire should be horizontal and pass freely in between the poles of magnet.
- The horse shoe magnet should be placed vertically at the center of the wire with its face normal to the length of wire.
- The current should not exceed one Ampere to avoid the overheating of the wire.
- The movement of bridges on the wire should be slow so that the resonance point can be found easily
- The diameter of the wire must be measured accurately at different points in two mutually perpendicular directions.
- The sonometer wire and the clamp used to hold the magnet should be non-magnetic.

SAMPLE ORAL QUESTIONS:

- What do you understand by the frequency of AC Mains?
- Distinguish between AC and DC. What is the use of magnet here?
- How does the sonometer wire vibrate when AC is passed through it?
- If you pass a DC through the wire, will it vibrate?
- What are the chief sources of errors in this experiment?
- What is the use of magnet here?
- What is Fleming's left hand rule?
- What is resonance?
- What is fundamental mode of vibration?
- Why do we take the material of wire to be non-magnetic?
- What is the principle of this experiment?

Physics Laboratory Manual

EXPERIMENTS NO. 02

Aim: To determine the wavelength of sodium lines by Newton's rings method.

Apparatus Required: Optical arrangement for Newton's rings, an optically plane glass plate, Convex lens, Sodium light source, Traveling microscope, magnifying lens, reading lamp and spherometer

Description of Apparatus:

The experimental apparatus for obtaining the Newton's rings is shown in the figure 1. A Plano-convex lens L of large radius of curvature is placed with its convex surface in contact with a plane glass plate P. At a suitable height over this combination, is mounted a plane glass plate G inclined at an angle of 45 degrees with the vertical. This arrangement is contained in a wooden box. Light from a broad monochromatic sodium source rendered parallel with the help of convex lens L_1 is allowed to fall over the plate G, which partially reflects the light in the downward direction. The reflected light falls normally on the air film enclosed between the Plano-convex lens L, and the glass plate P. The light reflected from the upper and lower surfaces of the air film produce interference fringes. At the center the lens is in contact with the glass plate and the thickness of the air film is zero. The center will be dark as a phase change of π radians is introduced due to reflection at the lower surface of the air film (as the refractive index of glass plate P ($\mu = 1.5$)) is higher than that of the air film ($\mu = 1$). So this is a case of reflection by the denser medium. As we proceed outwards from the center the thickness of the air film gradually increases being the same all along the circle with center at the point of contact. Hence the fringes produced are concentric, and are localized in the air film (figure 2). The fringes may be viewed by means of a low power microscope (travelling microscope) shown in the Fig. 1.

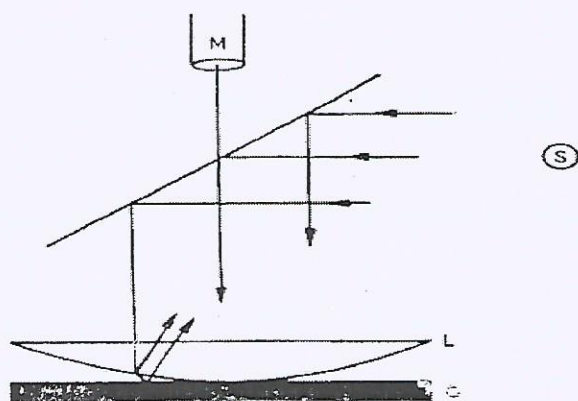


Fig 1

Fig. 1: Schematic representation of the experimental set up

Working Principle:

When a Plano-convex lens of large radius of curvature is placed with its convex surface in contact with a plane glass plate P a thin wedge shaped film of air is enclosed between the two. The thickness of the film at the point of contact is zero and gradually increases as we proceed away from the point of contact towards the periphery of the lens. The air film thus possesses a radial symmetry about the point of contact. The curves of equal thickness of the film will, therefore, be concentric circles with point of contact as the center (Fig.2).

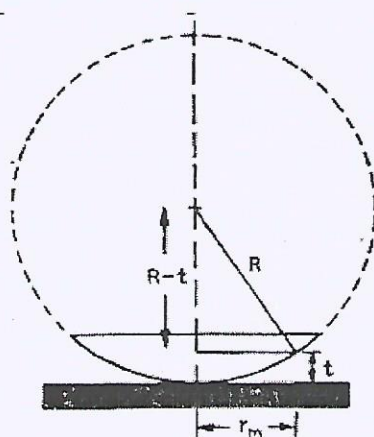


Fig. 2

Fig. 2: Schematic showing the radius of the m th dark ring for an air film thickness, t

Physics Laboratory Manual

In Fig. 3 the rays BC and DE are the two interfering rays corresponding to an incident ray AB. As Newton's rings are observed in the reflected light, the effective path difference x between the two interfering rays is given by:

$$x = 2\mu t \cos(t + \theta) + \lambda/2 \quad (10.1)$$

Where t is the thickness of the air film at B and θ is the angle of film at that point. Since the radius of curvature of the Plano-convex lens is very large, the angle θ is extremely small and can be neglected. The term $\lambda/2$ corresponds to phase change of π radians introduced in the ray DE due to reflection at the denser medium (glass). For air the refractive index (μ) is unity and for normal incidence, angle of refraction is zero. So the path difference x becomes:

$$x = 2t + \lambda/2 \quad (10.2)$$

At the point of contact, the thickness of the film is zero, i.e., $t = 0$, So $x = \lambda/2$. And this is the condition for the minimum intensity. Hence the center of the Newton's rings is dark. Further, the two interfering rays BC and DE interfere constructively when the path difference between the two is given by

$$x = 2t + \frac{\lambda}{2} = 2n\frac{\lambda}{2} \quad (10.3)$$

$$2t = (2n - 1)\lambda/2 \text{ [Maxima]} \quad (10.4)$$

And they interfere destructively when the path difference

$$x = 2t + \frac{\lambda}{2} = (2n + 1)\frac{\lambda}{2} \text{ or } 2t = \frac{2n\lambda}{2} \text{ [Minima]} \quad (10.5)$$

From these equations it is clear that a maxima or minima of particular order n will occur for a given value of t . Since the thickness of the air film is constant for all points lying on a circle concentric with the point of contact, the interference fringes are concentric circles. These are also known as fringes of equal thickness

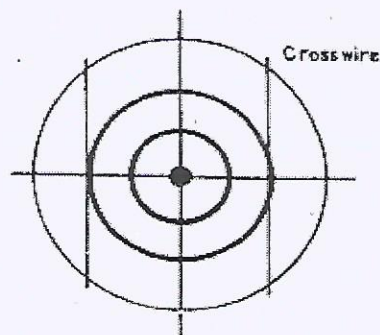


Fig. 3: Schematic representation of Newton's rings and the measurement of its diameter with cross wire.

Experimental Methods:

Calculation of diameters rings:

Let r_n be the radius of Newton's ring corresponding to a point B, where the thickness of the film is t , let R be the radius of curvature of the surface of the lens in contact with the glass plate P, then from the triangle CMB (figure 4), we have:

$$R^2 = r_n^2 + (R - t)^2 \text{ or } r_n^2 = 2Rt - t^2 \quad (10.6)$$

Since t is small as compared to R , we can neglect t^2 and therefore

$$R_n^2 = 2Rt, \text{ or } 2t = r_n^2 / R \quad (10.7)$$

If the point B lies over the n^{th} dark ring then substituting the value of $2t$ from equation (4) we have,

$$\left[\frac{r_n^2}{R} = \frac{2n\lambda}{2} \right], \text{ or } r_n^2 = n\lambda R \quad (10.8)$$

If D_n is the diameter of the ring then,

$$D_n^2 = 4n\lambda R \quad (10.9)$$

Similarly, if the point B lies over a n^{th} order bright ring we have

Physics Laboratory Manual

$$D_n^2 = 2(2n - 1)\lambda R \quad (10.10)$$

Calculation of λ :

From equation (7), if D_{n+p} is the diameter of $(n + p)^{th}$ bright ring, we have

$$D_{n+p}^2 = 2[2(n + 2) - 1]\lambda R \quad (10.11)$$

Subtracting equation (7), from equation (8), we get:

$$D_{n+p}^2 - D_n^2 = 4p\lambda R \quad (10.12)$$

$$\lambda = \frac{D_{n+p}^2 - D_n^2}{4pR} \quad (10.13)$$

By measuring the diameters of the various bright rings and the radius of curvature of the plano convex lens, we can calculate λ from the equation 9.

Formula used

The wavelength λ of the sodium light employed for Newton's rings experiments is given by:

$$\lambda = \frac{D_{n+p}^2 - D_n^2}{4pR}$$

Where D_{n+p} and D_n are the diameter of $(n + p)^{th}$ and n^{th} bright rings respectively, p being an integer number. R is the radius of curvature of the convex surface of the Plano-convex lens.

Methodology

1. Level the travelling microscope table and set the microscope tube in a vertical position. Find the vernier constant (least count) of the horizontal scale of the traveling microscope.
2. Clean the surface of the glass plate p , the lens L and the glass plate G . Place them in position as shown in Figure 1 and as discussed in the description of apparatus. Place the arrangement

Physics Laboratory Manual

in front of a sodium lamp so that the height of the center of the glass plate G is the same as that of the center of the sodium lamp. Place the sodium lamp in a wooden box having a hole such that the light coming out from the hole in the wooden box may fall on the Newton's rings apparatus and adjust the lens L_1 in between of the hole in wooden box and Newton's rings apparatus and adjust the lens L_1 position such that a parallel beam of monochromatic sodium lamp light is made to fall on the glass plate G at an angle of degrees.

3. Adjust the position of the travelling microscope so that it lies vertically above the center of lens L. Focus the microscope, so that alternate dark and bright rings are clearly visible.
4. Adjust the position of the travelling microscope till the point of intersection of the cross wires (attached in the microscope eyepiece) coincides with the center of the ring system and one of the cross-wires is perpendicular to the horizontal scale of microscope.
5. Slide the microscope to the left till the cross-wire lies tangentially at the center of the 20th dark ring (see Figure). Note the reading on the vernier scale of the microscope. Slide the microscope backward with the help of the slow motion screw and note the readings when the cross-wire lies tangentially at the center of the 18th, 16th, 14th, 12th, 10th, 8th, 6th and 4th dark rings respectively [Observations of first few rings from the center are generally not taken because it is difficult to adjust the cross-wire in the middle of these rings owing their large width].
6. Keep on sliding the microscope to the right and note the reading when the cross-wire again lies tangentially at the center of the 4th, 6th, 8th, 10th, 12th, 14th, 16th, 18th, 20th dark rings respectively.
7. Remove the plano-convex lens L and find the radius of curvature of the surface of the lens in contact with the glass plate P accurately using a spherometer. The formula to be used is :

$$R = \frac{l^2}{6h} + \frac{h}{2} \quad (10.14)$$

Where l is the mean distance between the two legs of the spherometer h is the maximum height of the convex surface of the lens from the plane surface.

8. Find the diameter of each ring from the difference of the observations taken on the left and right side of its center. Plot a graph between the number of the rings on X-axis and the

Physics Laboratory Manual

square of the corresponding ring diameter on Y-axis. It should be a straight line as given by the equation 9 (see figure). Taken any two points on this line and find the corresponding values of $(D_{n+2}^2 - D_n^2)$ and p for them.

9. Finally calculate the value of wavelength of the sodium light source using the formula.

Observations

Determination of the Least Count:

Determination of the Least Count of the Horizontal Scale of travelling Microscope

1. Value of one division of the horizontal main scale of travelling microscope = cm
2. Total number of divisions on the Vernier scale = which are equal to division of main scale of the Vernier scale =cm
3. Value of one division of the Vernier scale =cm
4. Least count of the horizontal scale of the microscope (given by the value of one division of main scale – the value of one division of Vernier scale) =
5. Pitch of the screw =cm
6. Number of division on circular head =
7. Least count of the spherometer =cm
8. Mean distance between the two legs of the spherometer, l =cm
9. The radius of curvature R of the Plano convex lens is (as given by equation 10):
$$R = [l^2 / (6h + h) / 2] = \text{.....cm}$$
10. The wavelength λ of sodium light is (as given by equation 9):

$$\lambda = \frac{D_{n+2}^2 - D_n^2}{4pR}$$

Calculations from the graph:

1. Plot a graph taking squares of the diameters, D_n^2 along the Y-axis and the number of rings along the X-axis.

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2. The curve should be a straight line.
3. Take two points P_1 and P_2 on this line and find the corresponding values of $(D_{n+2}^2 - D_n^2)$ and p from it, calculate the value of wavelength of the sodium light from these values.

Table 1: Determination of $(D_{n+2}^2 - D_n^2)$ and p

Order of The Rings	Reading of the microscope left hand side (a) cm	Right hand side (b) cm	Diameter of the ring ($a \sim b$) cm	Diameter $(a \sim b)^2 \text{ cm}^2$	$(D_{n+2}^2 - D_n^2)$ For $p=4 \text{ cm}^2$	Mean value of $(D_{n+2}^2 - D_n^2)$ For $p=4 \text{ cm}^2$
20					$D_{20}^2 - D_{16}^2 = .$	
18					$D_{18}^2 - D_{14}^2 = .$	
16					$D_{16}^2 - D_{12}^2 = .$	
14					$D_{14}^2 - D_{10}^2 = .$	
12					$D_{12}^2 - D_8^2 = .$	
10					$D_{10}^2 - D_6^2 = .$	
8					$D_8^2 - D_4^2 = .$	

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Table 2: Determination of R (radius of curvature of the lens L) using a spherometer

Sl No	Spherometer reading on		H ($a \sim b$) cm	Mean h cm
	Plane glass plate a cm	Convex surface of lens b cm		

Results:

The value of the wavelength of the sodium light as calculated

1. Using the observations directly =A°
2. Using the graphical calculations = A°
3. Mean value of the wavelength of sodium light = A°
4. Standard average value of the wavelength of the sodium light = 5893 A°
5. Percentage of error = %

Source of errors and precautions:

- The optical arrangements as shown in Figure 1 should be very clean (use spirit for cleaning these optical elements) and so made that the beam of light falls normally on the plano-convex lens L and glass plate P combinations.
- The plano-convex lens for the production of Newton's rings should have large value of radius of curvature. This will keep the angle of wedge shape air film very small and therefore the rings will have a larger diameter and consequently the accuracy in the measurements of the diameter of the rings will be increased.
- To avoid any backlash error, the micrometer screw of the travelling microscope should be moved very slowly and moved in one direction while taking observations.
- While measuring diameters, the microscope cross-wire should be adjusted in the middle of the ring.

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- The amount of light from the sodium light should be adjusted for maximum visibility. Too much light increases the general illumination and decrease the contrast between bright and dark rings.

Sample oral questions:

1. What do you understand by the interference of light?
2. What are essential conditions for obtaining interference of light?
3. What do you understand by coherent source?
4. Is it possible to observe interference pattern by having two independent sources such as two candles?
5. Why should be two sources be monochromatic?
6. Why are the Newton's rings circular?
7. Why is central ring dark?
8. Where are these rings formed?
9. Sometimes these rings are elliptical or distorted, why?
10. What is the difference between the rings observed by reflected light and those observed by transmitted light?
11. What will happen if the glass plate is silvered on the front surface?
12. What will happen when a little water is introduced in between the plano-convex lens and the plate?
13. How does the diameter of rings change on the introduction of liquid?
14. Can you find out the refractive index of a liquid by this experiment?
15. Is it possible to have interference with a lens of small focal length?
16. What will happen if the lens is cylindrical?
17. Why do the rings get closer and finer as we move away from the center?

Experiment No. 3

Objective: To determine the resistance per unit length of a Carey Foster's bridge wire and then to find the resistivity of the material of a given wire.

Apparatus: A Carey Foster's bridge, a Leclanche cell, Weston galvanometer, a 1-ohm coil, a decimal -ohm box preferably dial pattern, sliding rheostat of a small resistance, a single way plug key, thick copper strips, a shunt wire and connecting wires.

Theory of Carey Foster bridge :

Carey Foster's bridge is specially suited for the comparison of two nearly equal resistances whose difference is less than the resistance of the bridge wire. As shown in fig. 1, two resistances X and Y to be compared are connected in the outer gaps of the bridge in series of the bridge wire. These two resistances together with the bridge wire from the two arms of the Wheatstone bridge. One composed of X plus a length of the bridge wire up to the balance point and the second composed of Y plus the rest of the bridge wire. The remaining two arms are formed by two nearly equal resistances P and Q , which are connected in the inner gaps of the bridge. If l_1 be the reading on the scale of the position of the null point, we have, from usual Wheatstone bridge principle

$$\frac{P}{Q} = \frac{X + \sigma(l_1 + \alpha)}{Y + \sigma(100 - l_1 + \beta)}$$

or
$$\frac{P}{Q} + 1 = \frac{X + Y + \sigma(100 + \alpha + \beta)}{Y + \sigma(100 - l_1 + \beta)} \quad \text{————— (1)}$$

where α and β in units of length of the bridge wire, are the end corrections at the left and right ends of the bridge wire respectively and σ is the resistance per unit length of the bridge wire. If now X and Y are interchanged and l_2 be the reading on the scale of the position of the new null point, we have

$$\frac{P}{Q} = \frac{Y + \sigma(l_2 + \alpha)}{X + \sigma(100 - l_2 + \beta)}$$

$$\frac{P}{Q} + 1 = \frac{X + Y + \sigma(100 + \alpha + \beta)}{X + \sigma(100 - l_2 + \beta)} \quad \text{————— (2)}$$

Comparing equations (1) and (2) we see that the fraction on the right hand side are equal and since their numerators are identical their denominators must also be equal. Hence equating the denominators of the right hand sides of equation (1) and (2), we have

$$\begin{aligned}
 Y + \sigma(100 - l_1 + \beta) &= X + \sigma(100 - l_2 + \beta) \\
 X - Y &= \sigma(l_2 - l_1)
 \end{aligned}
 \tag{3}$$

Thus the difference between the resistances X and Y can be obtained by determining the resistance of the bridge wire between the two null points.

Working Principle: Let two resistances P and Q of nearly equal values be connected in the inner gaps of a Carey Foster's bridge and let a known resistance R be connected in the outer left gap of the bridge. Let a thick copper strip be connected in the outer right gap of the bridge and assume that l_1 and l_2 are respectively the reading on the scale of the positions of the null point on the bridge wire before and after interchanging the known resistance R and the thick copper strip in the outer gaps, then we have from eq. (3) by putting $X = R$, $Y = 0$.

$$R = \sigma(l_2 - l_1)$$

$$\sigma = \frac{R}{l_2 - l_1} \tag{4}$$

Now let the coil of unknown resistance X be connected in the outer left gap and a standard known resistance Y of nearly the same value in the outer right gap of the bridge. Then if l_1' and l_2' be the readings on the scale of the positions of the null point before and after interchanging X and Y , we have, from equation 3.

$$X - Y = \sigma(l_2' - l_1')$$

Where

$$X = \sigma(l_2' - l_1') + Y \tag{5}$$

This equation can be used to calculate X , if σ is determined from equation (4). Let L be the length of the wire and 'a' is the cross-sectional area of the wire. Then the resistivity ρ is related to X by

$$X = \rho \frac{L}{a} \tag{6}$$

$$\text{or } \rho = \frac{Xa}{L} \text{ ohm.cm} \tag{7}$$

Methodology of Determination of σ :

Connect a standard 1 ohm resistance in the left gap of a Carey Foster's bridge and a thick copper strip in its outer right gap as shown in Fig. Next connect the two resistance boxes P & Q at the inner gap of the C-F bridge as shown in the fig. Jockey is connected through the galvanometer as shown. Finally connect the Leclanche cell between A and C including a plug key K in the circuit.

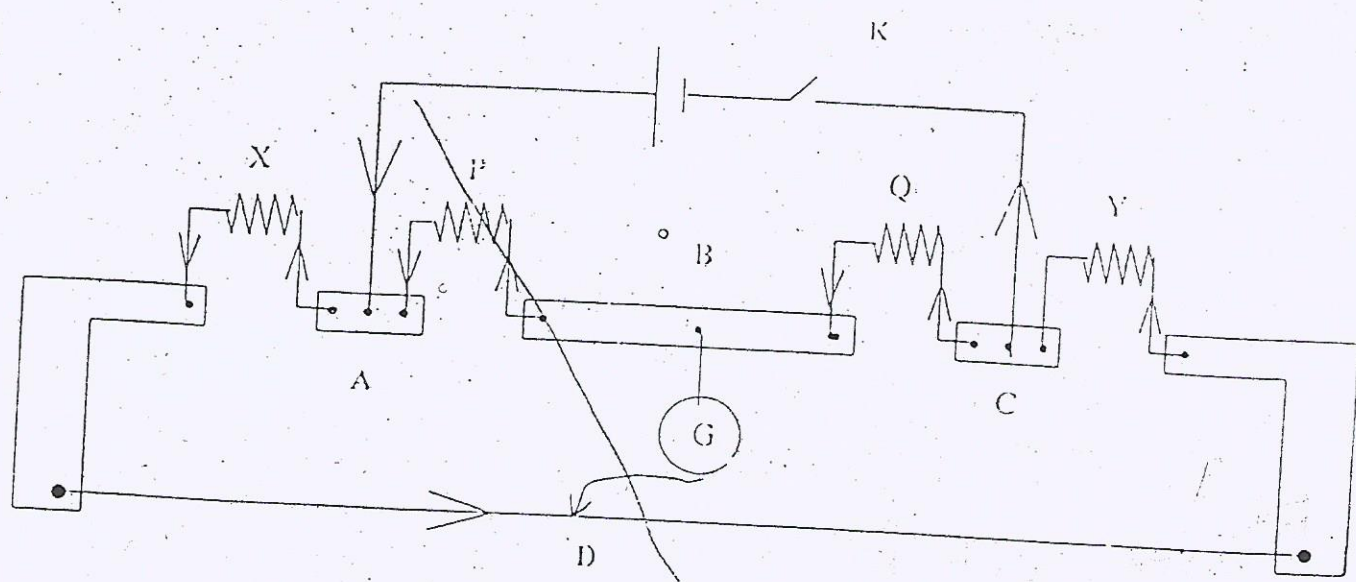
Put $P=Q$ from the resistance boxes and adjust for the null point. Measure l_1 and l_2 interchanging the two resistances in the outer gaps of the bridge. Follow the observation table. Calculate the value of σ for each set of observations separately from equation (4) and then find the mean value of σ .

To find the resistance of the given wire, replace the copper strip by the wire and repeat the process. Find the resistance of the given wire using equation 5 after measuring its radius and length of the wire. Calculate the resistivity using equation 7.

Sources of error and precautions:

- The ends of the connecting wires should be clean and all connections should be firmly made. The decimal - ohm box and thick copper strips should connect the given one-ohm resistance coil.
- A rheostat should be used to introduce the resistances P and Q in the inner gaps of the bridges and the sliding contact should be adjusted to be approximately in the middle. It is not absolutely necessary that P and Q should be exactly equal except for high sensitiveness of the bridge, nor should their values be known. If $P = Q$, the positions of null point before and after interchanging the resistances in the outer gaps will be at equal distances from the middle point of the bridge wire, provided, of course, the wire is uniform. If P and Q differ very much it will not be possible to obtain the two positions of the null point on the bridge wire. The use of rheostat to introduce P and Q in the inner gaps possesses several advantages. Besides being cheap, it is flexible, for it can be used to obtain the null point in any part of the bridge wire and also enables us to take several sets of readings for $(l_2 - l_1)$ for the same values of X and Y . With fixed values for P and Q this could not have been possible.
- In order that the bridge may have high sensitiveness, the resistances of the four arms should be of the same order.
- In order to reduce the inaccuracy in the result due to a small error in reading the position of the null point to minimum, the null points while comparing X and Y should lie as near the middle of the bridge wire as possible.
- While determining the value of ρ the value of R should be comparable with the resistance of the bridge wire so that the two positions of the null point before and after interchanging the resistances in the outer gaps lie near the ends of the bridge wire. The value of $(l_2 - l_1)$ will then be almost equal to the entire length of the bridge wire and the error in the value of σ due to non uniformity of the bridge wire will be reduced to minimum.
- A plug key should be included in the cell circuit and should be closed when observations are being made.

- The galvanometer should be shunted by a low resistance wire to avoid excessive deflection in it when the bridge is out of balance. The exact position of the null point should be determined with full galvanometer sensitivity by removing the shunt wire from it.
- The cell circuit should be closed before depressing the jockey over the bridge wire, but when breaking, reverse order should be followed.
- The jockey should always be pressed gently and the contact between the jockey and the bridge wire should not be made while the jockey is being moved along.



Carey Foster Bridge

OBSERVATIONS. (A) Determination of σ

S.N o	R Ohm	P=Q ohm	Position of balance point with copper strip in the		$(l_2 - l_1)$ cm.	σ ohm/cm	Mean σ in ohm/cm.
			Right gap l_1 cm	Left gap l_2 cm			

OBSERVATIONS. (B) Determination of resistance and resistivity of the unknown wire

S.No	R Ohm	P=Q ohm	Position of balance point with copper strip in the		$(l_2 - l_1)$ cm.	$X = R + \sigma(l_2 - l_1)$ ohm	X ohm mean.
			Right gap l_1 cm	Left gap l_2 cm			

OBSERVATION: (B) Determination of Resistivity:

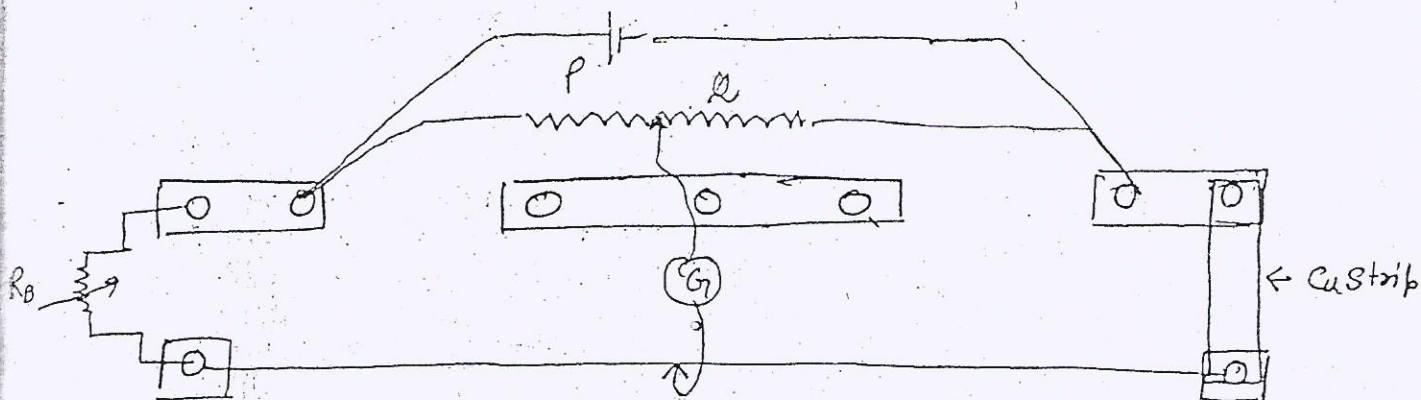
Length of the wire $L = \dots\dots\dots$ cm

Cross-sectional area of the wire $a = \dots\dots\dots$ cm²

$$X = \rho L/a$$

or $\rho = Xa/L$ ohm.cm.

$$= \dots\dots\dots \text{ohm.cm.}$$



Physics Laboratory Manual

EXPERIMENT NO. 04

Aim of the experiment: Measurement of voltage and frequency of a given signal using CRO.

Apparatus Required: Cathode Ray Oscilloscope, Function Generator(s), a pair of BNC Connectors

Introduction: A cathode ray oscilloscope (CRO) can be used to measure the voltage and frequency of given unknown signal. (A brief description of CRO is given at the end of this manual). A RC oscillator can be used to generate an electrical signal of desired frequency and amplitude. In the given experiment the RC oscillator has to be used to generate the signal and the CRO will be used to measure its voltage and frequency.

Procedure: Switch on the oscillator. Place the time base knob in horizontal input position and wait for a couple of minutes. Notice a bright spot of light on the screen of the CRO. The spot can be moved in vertical or horizontal direction by using the horizontal position knob and vertical position knob respectively. Place the time base in appropriate position (i.e. 1ms/cm or 0.1 ms/cm or any other value). A bright line can be noticed on the CRO screen. Your CRO is now ready to measure voltage and frequency of the unknown signal.

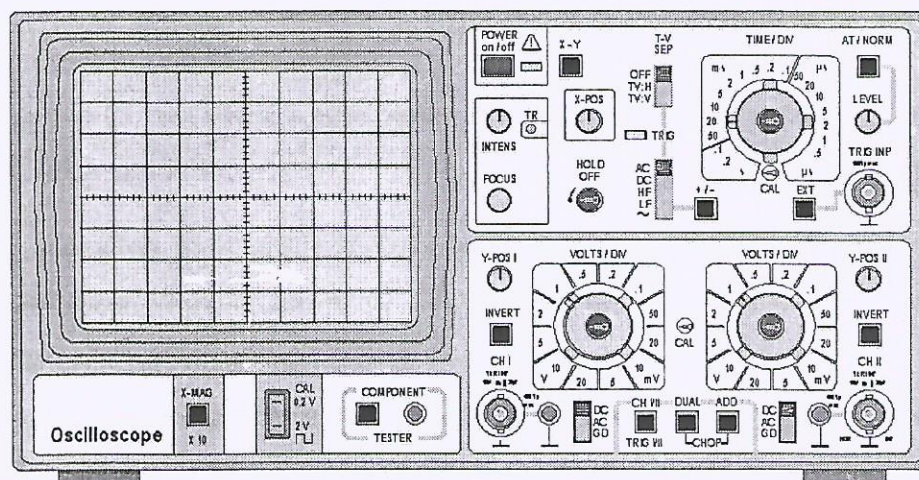


Fig. 1: The Cathode ray oscilloscope used in Physics Laboratory.

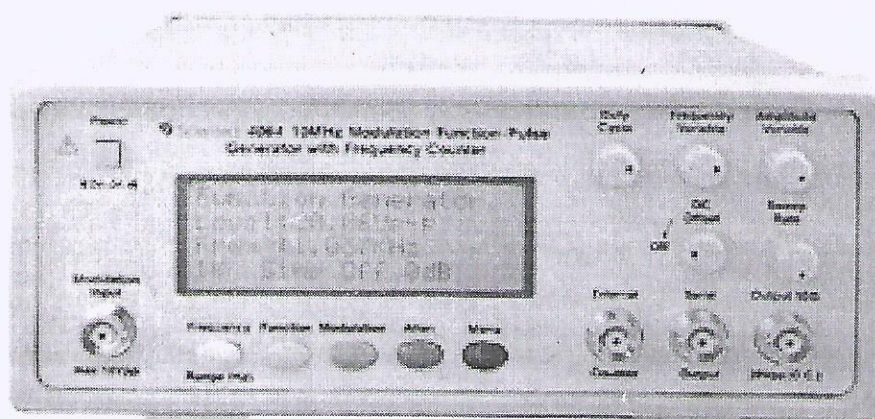


Fig. 2: The function generator used in Physics Laboratory.

Selection of frequency:

The RC oscillator is having several knobs, which can be used to select frequency of the signal to be generated. In the top left hand corner, you would see three knobs. These knobs can be used to select frequency value, which can be represented by three digits. For example, suppose you are setting the left knob to 6, the middle knob to 5 and the extreme right knob to 4. Then the selected frequency will be 654Hz. below these three knobs you will get a multiplier. The multiplier will multiply the above selected frequency. Thus if you select 654Hz and multiplier position is 10 then the overall frequency will be 6540Hz.

Selecting amplitude of the signal:

The RC oscillator provides you option to vary the voltage of the signal to be generated. This can be done using two voltage selecting knobs (Fine and Coarse). Therefore, using different knobs a signal of given amplitude and given frequency can be generated by the RC oscillator. This signal can now be used as an input to the CRO and its frequency and voltage can be measured.

Voltage Measurements:

Physics Laboratory Manual

Use the signal generated by RC oscillator as an input to CRO. Place the Y amplifier in proper value. From vertical scale measure the peak to peak value. This will give the value of peak to peak voltage of the signal.

Lissajous Figures:

Theory: When two simple harmonic motions are plotted against each other at right angles, the resulting configuration is called a Lissajous figure. Simple harmonic motions plotted against time gives sinusoidal configurations. Two sinusoidal electrical inputs given to an oscilloscope will give a Lissajous pattern on the screen. The particular pattern depends upon the frequency, amplitude and phase of the applied inputs. The frequency ratio of the inputs may be determined from an analysis of the Lissajous figure produced. If a Lissajous figure is enclosed in a rectangle whose sides are parallel to the formation axes of the figure, the frequency ratio of the two inputs may be determined by counting the points of tangency to the sides of the rectangle enclosing the pattern. Once the frequency ratio is known, the input frequency can also be determined from the same.

Procedure:

1. Connect one signal generator to the vertical input and the other to the horizontal input of the oscilloscope. Switch controls so that the oscilloscope accepts the output of the signal generator instead of the horizontal sweep. Set both the generators for 1000 cycles (say) and make gain adjustments until an ellipse of satisfactory size is observed on the screen. Adjust controls as necessary to stop the ellipse. By switching one of the generators off and on, cause the ellipse to change phase, noting the various shapes it assumes. By phase changes and amplitude adjustments, one may try to get a circular configuration.
2. Leaving the vertical input at 1000 cycles and assuming it to be the standard, adjust the horizontal input generator (the variable) approximately 500 c.p.s. to obtain the 1-2 Lissajous figure, a figure 8 on its side.
3. Next obtain the 2:1 pattern by varying the horizontal input frequency. This is an upright figure 8.

Physics Laboratory Manual

4. In like manner, obtain Lissajous figures down to 1:5 and upto 5:1. Sketch all the figures obtained and compare the frequency. Obtained from the Lissajous ratios with the dial reading of the horizontal input signal generator.

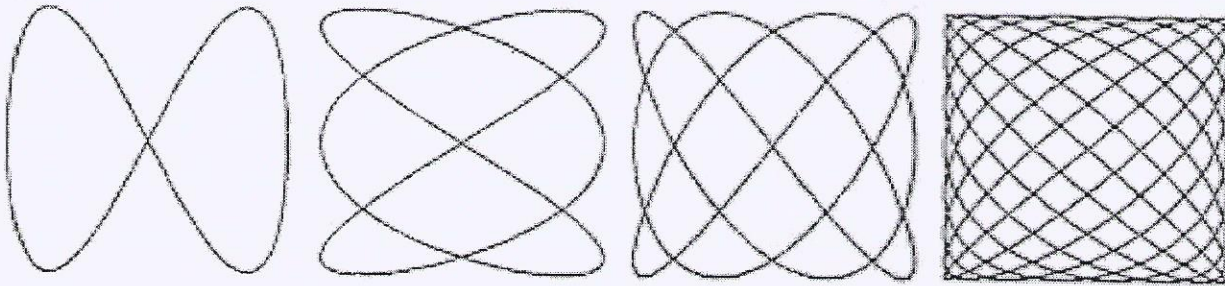


Figure 3: Sample Lissajous Figures

5. Now, remove one signal generator from the oscilloscope and connect the given unknown source. Changing the frequency of the signal generator, various Lissajous figures may be obtained (e.g. circle, 8 shape, etc.). Hence, from the known ratio of the respective Lissajous figures, the frequency of the AC source can be measured.

Observation Table:

Table 1: Measurement of voltage by CRO

Sl. No.	Voltage Source	from	RMS voltage V_{rms} measured by FG (V)			Ratio of V_{p-p}/V_{rms}
			Y-amplifi setting (V/div)	Vertical scale No of div.	V_{p-p} (V)	

Table 2: Measurement of frequency

Sl. No.	Freq. of function generator (f in Hz)			Freq. measurement using CRO (f_o in Hz)			Ratio f/f_o
	Digits	Multiplier	f (in Hz)	Value of time base	No. of div.	f_o (in Hz)	

Physics Laboratory Manual

Table 2: Measurement of unknown frequency with Lissajous figures

Horizon input Frequency on dial	Shape of figure	No. of tangency Points on X-axis on Y-axis	Vertical/	Horizontal

Precautions: The precautions to be taken while performing the experiment are:

1. Correct output/input terminals of function generators and CRO should be chosen.
2. Make sure that correct types of wave output have been chosen from function generator (Sine or Triangular or Square or DC)
3. Don't crank up the voltage level from function generator at its maximum limit.
4. The image produced over the CRO screen should be sharp and as thin as possible to take the correct readings.

Physics Laboratory Manual

EXPERIMENT NO. 08 05

Aim of the Experiment: To determine the wavelength of prominent spectral lines of mercury light by a plane transmission grating using normal incidence.

Apparatus Required: A spectrometer, Mercury lamp, Transmission grating, Reading lamp and Reading lens.

Theory: Basic description of spectrometer is given in page no. 35.

The wavelength λ of any spectral line using plane transmission grating can be calculated from the formula $(e+d) \sin\theta = n\lambda$, where $(e+d)$ is the grating element, θ is the angle of diffraction, and n is the order of the spectrum. If there are N lines per inch ruled on the grating surface, then the grating element is given by $(e+d) = 2.54/N$ cm. Hence $(2.54/N) \sin\theta = n\lambda$, or $\lambda = 2.54 \sin\theta / nN$ cm.

Procedure

1) Adjustment of the grating for normal incidence:

The initial adjustment of the spectrometer is made as usual. The plane transmission grating is mounted on the prism table. The telescope is released and placed in front of the collimator. The direct reading is taken after making the vertical cross-wire to coincide with the fixed edge of the image of the slit, which is illuminated, by a monochromatic source of light. The telescope is then rotated by an angle 90° (either left or right side) and fixed. The grating table is rotated until on seeing through the telescope the reflected image of the slit coincides with the vertical cross-wire. This is possible only if the reflected image of the slit coincides with the vertical cross-wire. This is possible only when a light emerging out from the collimator is incident at an angle 45° to the normal to the grating. The vernier table is now released and rotated by an angle 45° towards the collimator. Now light coming out from the collimator will be incident normally on the grating. (Fig 1).

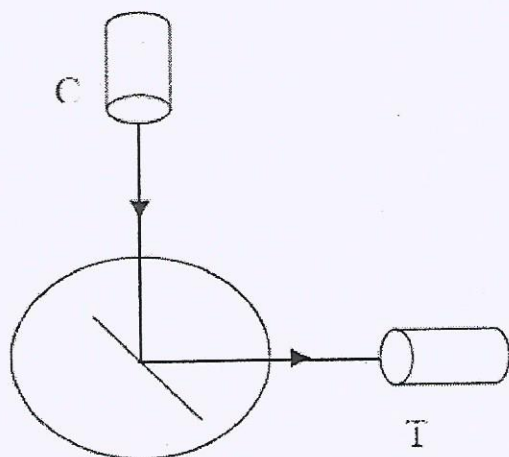


FIG.1

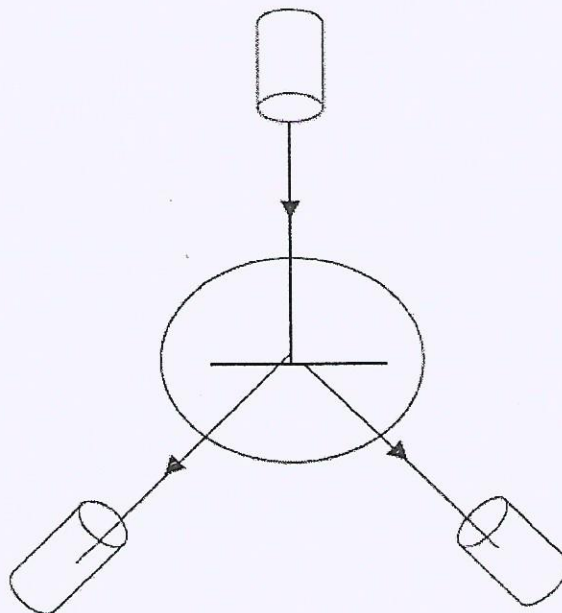


FIG.2

2) Standardization of the grating:

The slit is illuminated by sodium light. The telescope is released to catch the different image of the first order on the left side of the central direct image. The readings in the two vernier are noted. It is then rotated to the right side to catch the different image of the first order, the readings are noted. (Fig.2). The difference between the positions of the right and left sides given twice the angle of diffraction $2\theta^\circ$. The number of lines per meter of the grating (N) is calculated by using the given formula assuming the wavelength of the sodium light as 589.3 nm .

3) Wavelengths of the spectral lines of the mercury spectrum:

The sodium light is removed and the slit is now illuminated by white light from mercury vapour lamp. The telescope is moved to either side of the central direct image, the diffraction pattern of the spectrum of the first order and second order are seen. The readings are taken by coinciding the prominent lines namely violet, blue, bluish – green, yellow1, yellow2 and red with the vertical wire. The readings are tabulated and from this, the angles of diffraction for different colours are

Physics Laboratory Manual

determined. The values are tabulated in Table 1. The wavelengths for different lines are calculated by using the given formula.

6. Observations:

A. For the adjustment of grating for normal incidence:

1. Least count of the Spectrometer scale:

Value of 1 division of main scale =

Divisions of main scale are equal to divisions of vernier scale.

Value of 1 division of vernier scale.

Least count of Spectrometer scale:

= value of 1 division of main scale – 1 division of vernier scale.

2. Reading of the telescope for direct image of the slit:

V1 = V2 =

3. Reading of the telescope after rotating it through 90° :

V1 = V2 =

4. Reading of prism table when reflected image of the slit coincides with the vertical cross wire V1 = V2 =

1. Reading of prism table when rotated through 45° or 135° :

V1 = V2 =

1. Number of lines N ruled per inch on the grating

2. Grating element $(e+d) = 2.54/N$

B. Calculations:

For first order, $n=1$, $\lambda = (e+d) \sin\theta/2$ cm.

λ for Violet I colour =cm.

Calculate λ for all visible spectral lines also.

Physics Laboratory Manual

Table 1: Table for the measurement of the angle of diffraction θ

Order	Color of the spectral line	Spectrum to the left of the direct images			Spectrum to the right of the direct images			$2\theta = X - Y$	Angle $= \theta$
		M.S.R	V.S.R	T.R=(X)	M.S.R	V.S.R	T.R=(Y)		
1 st order	Violet Window 1 Window 2								
	Blue Window 1 Window 2								
	Green Window 1 Window 2								
	Yellow Window 1 Window 2								
	Red Window 1 Window 2								
2 nd order	Violet Window 1 Window 2								
	Blue Window 1 Window 2								
	Green Window 1 Window 2								
	Yellow Window 1 Window 2								
	Red Window 1 Window 2								

Physics Laboratory Manual

Results: The experimental values of the wavelength of spectrum colours along with the percentage errors are finally tabulated in Table 2.

Table 2: Experimental values and verification of wavelengths of spectrum colours.

Colour of the Spectral line	Wavelength as obtained by experiment	Standard value of wavelength	Percentage error %
Violet I		4047 Angstroms	
Blue		4358 Angstroms	
Green		5461 Angstroms	
Yellow I		5770 Angstroms	
Red		6234 Angstroms	

7. Source of Error and Precautions

1. The axes of the telescope and the collimator must intersect at and be perpendicular to the main axis of the spectrometer.
2. The collimator must be so adjusted as to give out parallel rays.
3. The telescope must be so adjusted as to receive parallel rays and form a well-defined image of the slit on the crosswire.
4. The prism table must be optically leveled.
5. The grating should be so mounted on the prism table that its ruled lines are parallel to the main axis of the spectrometer.
6. The plane of the grating should be normal to the incident light and its ruled surface must face the telescope so that the error due to non-parallelism of the incident rays is minimum.
7. The slit should be as narrow as possible and parallel to the ruled surface of the grating.
8. While handling the grating one should not touch its faces but hold it between the thumb and the fingers by edges only.
9. While taking observations of the spectral lines, the prism table must remain clamped.
10. The reading of both the verniers should be recorded. This eliminates the error due to non-coincidence of the center of the graduated scale with the main axis of the spectrometer.

Physics Laboratory Manual

8. Sample Questions:

1. What do you understand by diffraction of light?
2. How does it differ from interference of light?
3. What is a diffraction grating?
4. How do you measure the wavelength of light using grating?
5. What is grating element?
6. How do you adjust telescope and collimator for parallel rays?
7. How do you set the grating for normal incidence?
8. Why the ruled surface of grating face should face towards the telescope?
9. How many orders of spectra are you getting with the grating?
10. What is the difference between a prism spectrum and a grating spectrum?
11. What are the various series of lines observed in hydrogen spectrum?
12. What is Rydberg constant?
13. A transmission grating with 2000 lines/cm is illuminated by a beam of 694.3 nm light from a laser. Spots of light, on both sides of the undeflected beam, appear on a screen 2.0 m away.
 - ❖ How far from the central axis is either of the two nearest spots?
 - ❖ Find how much difference it makes whether you use the approximation $\sin \theta \approx \theta$

Experiment No. 6

Objective: To calibrate the Voltmeter using a potentiometer

Apparatus required:

A potentiometer, two storage batteries, two rheostats, a standard Leclanche cell, a Galvanometer, a voltmeter to be calibrated, a two-way key, two single way keys and connecting wires.

Theory:

In Fig.1, potentiometer has been represented by the line AB, where a constant potential difference has been maintained between the points A and B. The potential of A is higher than the potential at the point B. Let the positive pole of a Leclanche cell D be connected to the end A and the negative through a galvanometer to the jockey of the potentiometer. Then if l_1 is the length of the potentiometer wire for no deflection in the galvanometer, we have

$$E = k l_1$$

where E is the emf of the Leclanche cell and k is the potential gradient of the potentiometer. This gives the value of k.

Now let the Leclanche cell be replaced by all portion MP of the rheostat MN through whole of which a steady current is maintained, the higher potential point M of the rheostat being connected to the end A of the potentiometer. Then if l_2 is the length of the potentiometer wire at the null point, the PD across the portion MP of the rheostat is given by, $V = k l_2$

If the PD across the same portion MP of the rheostat as measured by a voltmeter is V' , the error in the reading V of the voltmeter is $(V' - V)$. Now if the PD between the points M and P of the rheostats is made variable by making one of the points, say P variable and the errors corresponding to the various readings of the voltmeter thus obtained be determined by the potentiometer, a calibration curve of the voltmeter may be plotted by taking V' as abscise and $(V' - V)$ as ordinates.

Methodology :

Complete the circuit as shown in Figure 1, with the following hints. Connect the two ends A and B of the potentiometer to a storage battery S1 in series with a rheostat R1, including a plug key K1 in the circuit. Connect the two fixed terminals M and N of a rheostat R2 to a battery S2 including a plug key K3 in the circuit. Connect one terminal of the voltmeter V to be calibrated through a key K2 to the higher potential fixed terminal M of the rheostat R2 and the second terminal of the voltmeter to the sliding terminal P of the rheostat.

Now connect the positive pole of the Leclanche cell and the higher potential fixed terminal M of the rheostats R2 to the point A of the potentiometer. Finally connect the negative pole of the Leclanche cell and sliding terminal P of the rheostat R2 through a two-way key and a galvanometer to the jockey Y which

slide:
the L
rheo
will g
Next
key
pote
R2.
calci
voltr

Alte
R2
reac
equ
erro

Use

(a)

(b)

Ler

At

exp

At

exp

slides over the potentiometer wire. Close the K1 and connect the negative pole of the Leclanche cell by means of the two way key to the galvanometer. Adjust the rheostat R1 in the main circuit till there is no deflection in the galvanometer. This will give the potential gradient along the potentiometer slide wire.

Next connect the sliding terminal P of the rheostat R2 by means of the two-way key to the galvanometer and determine the equivalent length l_2 of the potentiometer wire corresponding to the PD across the portion MP of the rheostat R2. Note down the reading V' of the voltmeter, open the keys K1 and K2 and calculate the true value V of the PD corresponding to the reading V' of the voltmeter and then the error $(V'-V)$ in the voltmeter reading.

Alter the PD between M and P by moving the sliding terminal P of the rheostat R2 and determine as before, the error $(V'-V)$ corresponding to the various readings V' of the voltmeter thus obtained covering the entire range of the equipment. Finally taking voltmeter readings as abscise and corresponding errors as ordinates, plot the calibration curve of the voltmeter.

Useful Hints:

- The magnitude of PD between A and B must be greater than the maximum PD to be measured by means of the potentiometer.
- Check the emf of the battery S2 is greater than the range of the voltmeter to be calibrated.
- The contact between the jockey and the potentiometer wire should be momentary and should not be made while the jockey is being moved along.
- The voltmeter should be calibrated over its entire range.

Tables

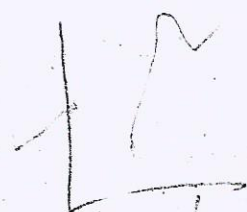
(a) Standardization of potentiometer wire

(b) EMF of standard Leclanche cell = 1.4 V

Length of potentiometer wire corresponding to EMF of Leclanche cell.

	Total length of slide wire div.	Equivalent length l_1 div
At start of experiment		
At the end of the experiment		

$\text{err } \% = \frac{355}{1000} \times 100 = 35.5\%$



(c) Calibration of voltmeter
Least count of voltmeter = Volts

Sl.No	Length of potentiometer wire corresponding to PD across the portion MP of the rheostat R2	Accurate value of PD V Volts	Reading of voltmeter V' Volts	Error in the reading of voltmeter (V' - V) Volts

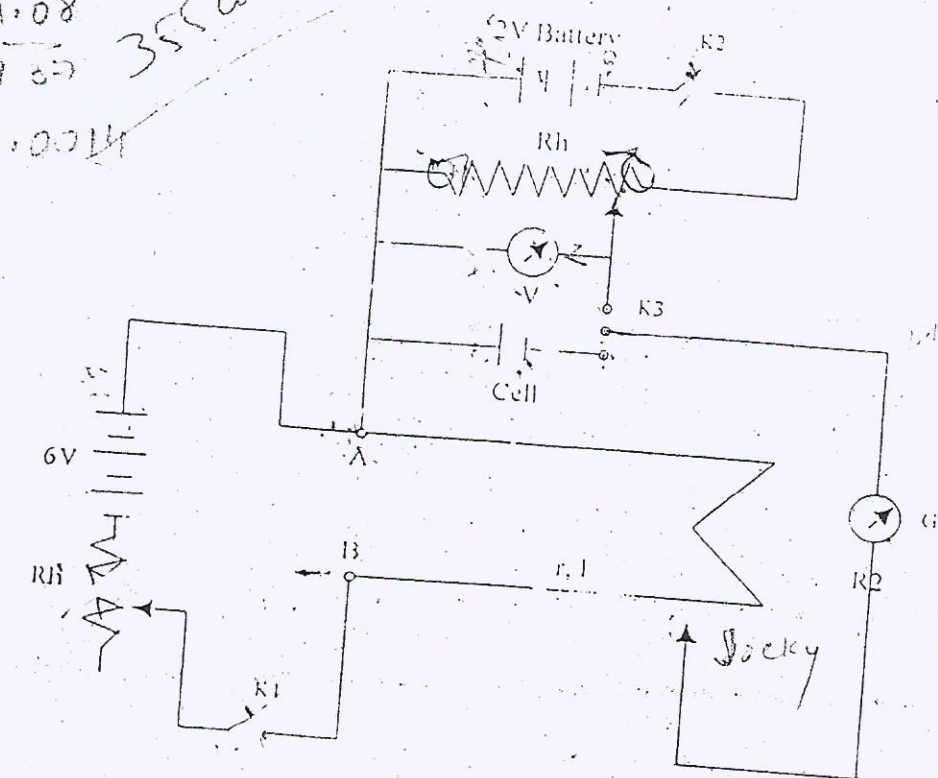
Results:

- The circuit has been successfully completed and checked.
- The graph so obtained by plotting the errors against the voltmeter reading is the calibration curve of the given voltmeter.

$K = 1.08$

$\frac{355}{1000}$

355 cm



Circuit diagram for the calibration of voltmeter using potentiometer

212

51

Se
var
volt
sigr
osc
and

Physics Laboratory Manual

EXPERIMENT NO. 07

Objective: To study LR circuit with an a.c. and determine the inductance of an inductor, its internal resistance and power factor of the circuit.

Equipment: Inductive Coil, known resistance, ammeter, variable ac supply, voltmeters, connecting wires.

Working principle: A series LR is connected across an a.c. supply as shown in figure 1 and voltages V_0 , V_R , V_L and current I_0 are measured. If we have a pure inductor L , a non inductive resistance R is connected in series with an inductance and an ammeter and a.c. supply. There are three voltages, supply voltage V_0 , across R is V_R and voltages across L is V_L and are measured with the voltmeters. If we have a pure inductor L , then if the current in the circuit is $i \cos(\omega t)$, the e.m.f. across the inductor is given by

$$V_L = \omega L i_0 \cos(\omega t + \pi/2) \quad (1)$$

Meaning thereby that the voltage across it leads i and hence the voltage V_R across the resistor by $\pi/2$. But in actual practice, every inductor has some resistance (r) associated with this is called its d.c. resistance. Apart from these there are iron losses and copper losses which play a significant role in a.c.. These losses could be effectively represented by a series resistance, sometimes referred to as the a.c. resistance. Usually the series resistance r is to take into account the d.c. resistance as well the a.c. resistance of the inductor. Thus the vector diagram for the circuit is given in figure 1.

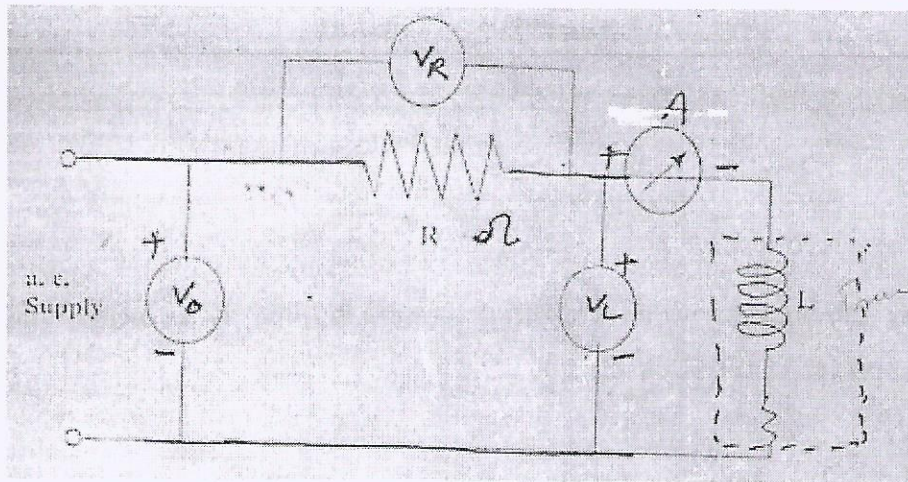


Figure 1. L-R Circuit Diagram

The voltage V_L does not lead the current by $\pi/2$. This voltage across the inductor can be seen to lead the voltage V_R across the resistor by an angle less than $\pi/2$. $\cos\Phi$ is the power factor of the inductive load. If a semi-circle is drawn on V_0 as diameter as shown in figure 2. OP produced cuts this at S. SQ then represents the p.d. across the inductive part (equal to ωLi) while PS represents the p.d. across the resistive part of the inductance V_L^r (equal to $r.i$) where r is the series resistance of the inductance (the sum of the d.c. and a.c resistance) from the vector diagram we have

$$V_0^2 = V_R^2 + V_L^2 + 2V_R V_L \cos\Phi \quad (2)$$

Neglecting the currents taken by the voltmeters across R and L the current in R is the same as the load current i hence $V_R = R.i$. then equation (2) becomes

$$V_0^2 = V_R^2 + V_L^2 + 2iR V_L \cos\Phi \quad (3)$$

If $W = iV_L \cos\Phi$ is the power consumption of the inductive load then (using eq.(3))

$$W = iV_L \cos\Phi = \frac{V_0^2 - V_R^2 - V_L^2}{2R} \quad (4)$$

Thus from the measurement of the three voltages and the resistance R we can calculate W and $\cos\Phi$. Further, if the load current i were measured with an ammeter, the apparent power consumption W , $\cos\Phi$ may also be computed from the relation

$$\cos\Phi = W / iV_L \quad (5)$$

Also from the vector diagram $PS = ir = V_r$, hence $r = V_r / I$ gives the series resistance of the inductor. Compare this with the d.c. resistance in inductor and account for the difference, from $QS = \omega Li = V_L$, we can calculate L . From

$$L = V_L / 2\pi f i \quad (6)$$

Methodology:

- Measure the non inductive resistance R with a multi meter and also the d.c. resistance of the inductor.
- Make the connections as shown in the figure. Adjust the variac, say 85 volts, and switch on the power. Take the readings of the voltmeters giving V_0 , V_R , and V_L . Measure the current i with the ammeter A . Care should be exercised in reading the voltmeters, where the pointer lies in between two graduations of a voltmeter scale, approximation must be made to read the fraction. This has to be done because the voltage occur in the second power in the expressions for W and $\cos\Phi$.
- Repeat the experiment by altering the input voltage V_0 with the variac in steps of 20 volts.
- For each set of observation, calculate W , $\cos\Phi$, i V_L and $\cos\Phi$ again from the relation $W/i V_L$.
- Draw vector diagram for each set of observation in the manner depicted in the figure. Measure Φ and calculate $\cos\Phi$ from the cosine tables.

From each of the vector diagrams measure the angle $\Phi' = QOP$ giving the phase difference between V_0 and V_R . Calculate the mean value and show that

$$\tan\Phi = L\omega / R + Lr$$

Where L and r have been determined from the vector diagrams, $\cos\Phi'$ power factor in the entire circuit.

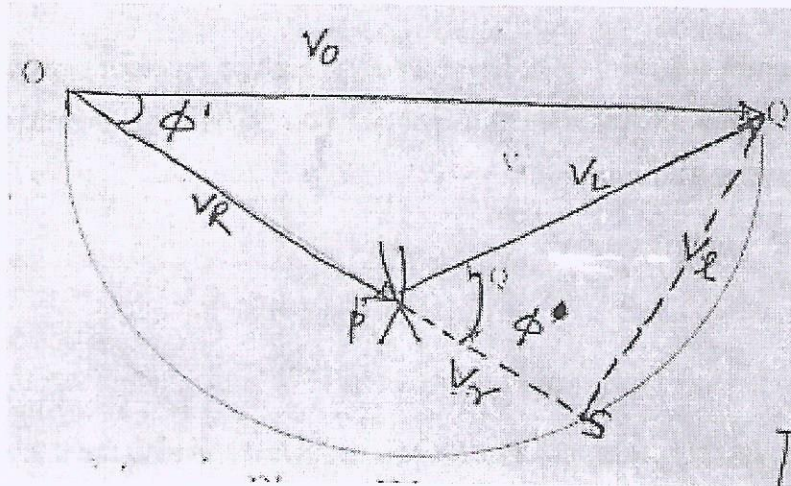


Figure 2. Phase Diagram

Observations:

P.D. in volts across

Sl. No	Ammeter reading (i) amp	L and R Combined V_0 volts	Across R V_R Volts	Across L V_L volts	$V_0^2 - V_R^2 - V_L^2$
1					
2					
3					
4					
5					
6					

Calculations:

$$W = \frac{V_0^2 - V_R^2 - V_L^2}{2R}$$

And

$$\cos\Phi = \frac{V_0^2 - V_R^2 - V_L^2}{2V_L V_R}$$

Make similar calculations and tabulate your results as given below:

Sl. No	W(True watts)	$\cos\Phi$	iV_L (apparent watts)	$\cos\Phi = W / iV_L$
1				
2				
3				
4				
5				
6				

Make similar calculation from other diagrams and tabulate your results as given below:

Sl. No	I Amp.	R in ohms	L in Henry	Φ	$\cos\Phi$
1					
2					
3					
4					
5					
6					

Results:-

(a)

$$\cos\Phi = \frac{V_0^2 - V_R^2 - V_L^2}{2V_L V_R} = \text{-----}$$

(b)

$$\cos\Phi = W / iV_L = \text{-----}$$

(c) $\cos\Phi$ (from vector diagram) =

(d) The series resistance r of the inductor =ohms

(d) The inductance of the inductor =Henry

Conclusion:

Sources of errors and precautions:

Physics Laboratory Manual

EXPERIMENT NO. 08

Objective: To study the single slit diffraction pattern and to measure the slit width using a laser.

Apparatus Required: He-Ne laser, Adjustable slit, Screen, Optical bench.

Theory: The phenomena of interference and diffraction are the manifestations of the wave nature of light. Diffraction is very often referred to as the bending of the waves around an obstacle. Whenever wave is limited in spatial extent by an aperture or an obstacle, diffraction takes place.

When the source and the object and the observation screen are both located at infinity with respect to the slit, the obtained diffraction pattern is known as Fraunhofer diffraction. When a collimated beam illuminates an aperture and the diffraction pattern is observed at infinity. Often a lens is inserted after the aperture and the diffraction pattern is studied at its focal plane.

Working principle: We consider a slit of width b that is infinitely long along y direction. It is illuminated by a collimated beam as shown in figure 1. The diffraction angle for the m^{th} minima is given by

$$\sin \theta_m = m\lambda / b$$

Where λ is the wave length of light.

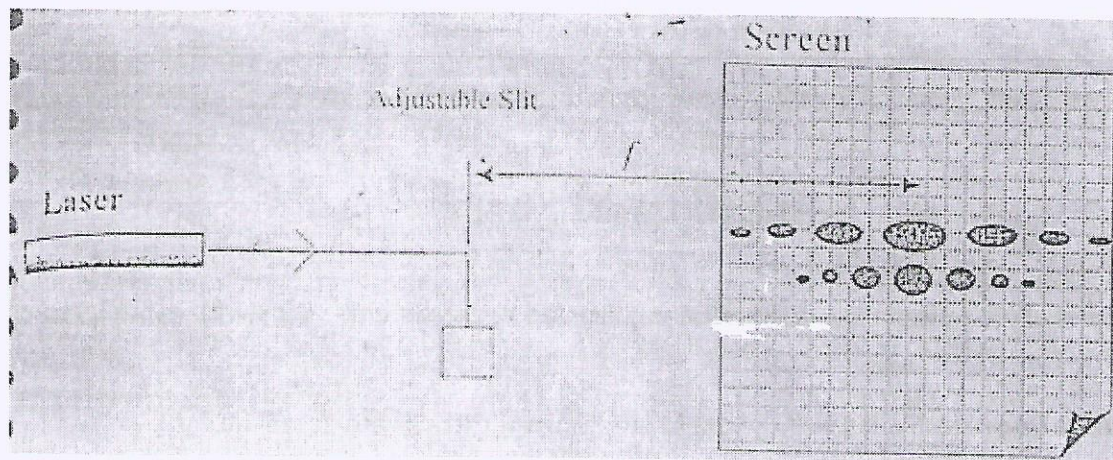


Figure 1. Schematic to observe Fraunhofer diffraction pattern from single slit.

The condition $m = 0$ gives the maximum irradiance that occurs in the direction $\theta_m = 0$. The angular separation between the consecutive minima is λ/b , the only exception being the first order minima about the central maximum which are λ/b apart. It can also be shown that secondary maxima are also λ/b apart. The central maximum, therefore has twice the width of the secondary maxima.

Experimental procedure: Align the laser with respect to the optical bench as well with the table top. Now mount the adjustable single slit and allow the laser to pass through. Adjust the slit to a minimum width and stop the laser light completely. Now record the initial reading of the micrometer screw, open the slit to a very small opening. Observe the diffraction pattern, record the micrometer screw reading and record the diffraction pattern on a graph sheet. Record the diffraction pattern on a graph sheet. Record the diffraction pattern for different slit widths. Record the position of diffraction maxima from the central maxima and plot the data.

The slit width can be calculated using the formula

$$b = \frac{m - m'}{X_m - X_{m'}} f \lambda \quad (1)$$

Where; X_m and $X_{m'}$ are respectively position of m and m' order minima on the screen, λ is the wavelength of laser light and f is the distance between slit and screen.

$$b = \frac{m - m'}{X_m - X_{m'}} \cdot f \cdot \lambda$$

Observation:

Micrometer initial reading:mm

Final reading:mm

Sl. No.	Order M	Distance from zeroth order X_m			$m - m'$	$X_m - X_{m'}$
		Left	Right	Mean		
1	0	0	0			
2	1					
3	2					
4	3					
5	4					
6	5					

Prepare independent table for different slit widths

Calculation: As per formula given in Equation (1), for different sets of observations.

Result: (b)
Calculated value of slit width is -----

(b)
Standard value of slit width is -----

Percentage Error:

Conclusion:

Sources of errors and precautions:

Physics Laboratory Manual

EXPERIMENT NO. 09

Objective

To determine the wavelength of a monochromatic source of light with the help of Fresnel's biprism.

Apparatus Required: Optical bench, Biprism, Slit, Micrometer eye-piece, Monochromatic source of light (sodium lamp), Convex lens

Formula Used

The wavelength (λ) of the sodium light is calculated by the formula in case of biprism experiment

$$\lambda = \beta d / D$$

Where: β = fringe width i.e., the distance between two successive maxima or minima of the fringes, d = distance between the two virtual sources (S_1S_2), D = distance between the slit and screen or eye-piece where the fringes are observed and measured.

Theory

Suppose that the monochromatic light from a slit S , perpendicular to the plane of the paper, falls symmetrically on the biprism, the refracting edge of which is perpendicular to the paper. The edge B of the biprism divides the incident wave-front into two parts which passing through the upper half and the lower-half of the biprism appear to diverge from the virtual images S_1 and S_2 . These two images now serve as the two coherent sources produced from the same source and are very close to and equidistant from S . Consequently, interference fringes are observed on the overlapping region of the two emergent beams of light (Figure 1).

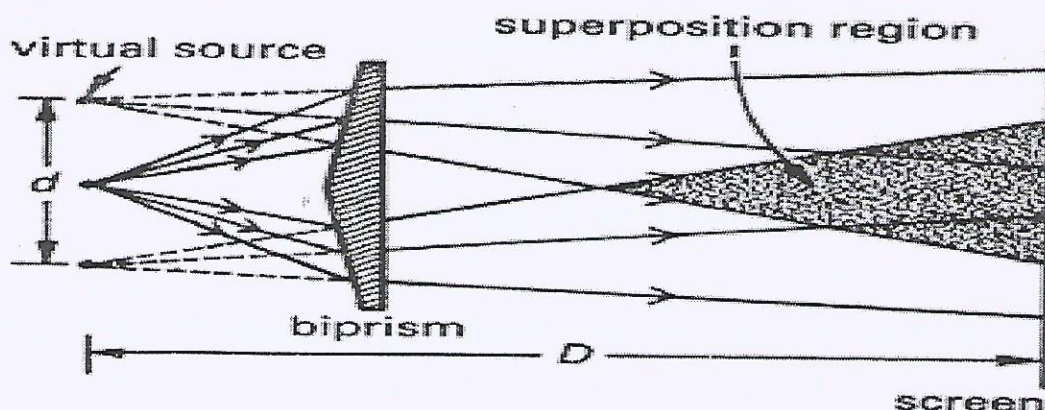


Figure 1. Formation of interference fringes from Fresnel's Biprism

The wavelength of monochromatic source of light can be calculated by the formula

$$\lambda = \beta d / D$$

Where β = fringe width i.e., the distance between two successive maxima or minima of the fringes, d = distance between the two virtual sources (S1S2), D = distance between the slit and screen or eye-piece where the fringes are observed and measured.

Procedure and Observations

1. The bed of the optical bench is leveled by a spirit level and the leveling screws.
2. Light source and slit is arranged in order to get the maximum light incident on the slit.
3. The centre of slit, biprism and eye piece is arranged at same height.
4. The slit and biprism edge are made vertical and in line parallel to the bench.
5. Observe the interference fringes with eye piece. For clear fringe pattern adjust the edge of biprism by rotating screw.
6. If the line joining the slit and the central edge of the biprism is not parallel to the length of the bench, fringes would shift laterally as the eye-piece is moved. To remove this, the biprism is moved a small distance transversely to the bench in a direction opposite to the direction of the shift till this lateral shift vanishes.

7. Now measure the position of fringes by putting the cross wire at mid position of each bright fringe.
8. For measurement of d , a lens of short focal length is inserted between the slit and the eye-piece.
9. The eye-piece is moved away from the slit so that the distance between the slit and the eye-piece is greater than four times the focal length of the convex lens used for the measurement of ' d '.

Measurement of β

The cross-wire is set at the centre of the first bright fringe and the reading of the micrometer screw is taken. The screw is then moved in one direction so that the wire falls in succession at the centers of the bright fringes and the corresponding readings are taken. From these readings, the width of a number of fringes (say, 10) are calculated by subtracting the first fringe from eleventh, second from twelfth and so on. After taking the mean, fringe width β for one fringe is calculated.

Measurement of d

A convex lens is mounted on the optical bench between the biprism and the eye-piece. The distances d_1 and d_2 between the well-defined images of the two virtual slits S_1 and S_2 are measured with the micrometer screw for the two positions of the lens. Then the distance between S_1 and S_2 is given by

$$d = \sqrt{d_1 d_2}$$

Measurement of D

The position of the slit and the eye-piece are noted. The distance between them gives observed value of D .

Table 1: [A] Reading for the measurement of Fringe Width (β)

Pitch of the screw =

No. of divs. on the circular head =

Least count of the micrometer screw =

No. of fringes (a)	Micrometer reading (in mm)	No. of fringes (b)	Micrometer reading (in mm)	Width of 10 fringes (in mm) (b-a)	Fringe Width (β)
1		11			
2		12			
3		13			
4		14			
5		15			
6		16			
7		17			
8		18			
9		19			
10		20			

[B] Readings for the determination of d

Position I			Position II			d = $\sqrt{d_1 d_2}$
Micrometer Reading of Image 1 in mm	Micrometer Reading of Image 2 in mm	Difference (d1)	Micrometer Reading of Image 1 in mm	Micrometer Reading of Image 2 in mm	Difference (d2)	

Result:

The wave length of the monochromatic source = ----- Å.

Percentage Error:

Precautions:

1. The setting of uprights at the same level is essential.
2. The slit should be vertical and narrow.
3. Crosswire should be fixed in the center of the fringe while taking observations for fringe width.
4. The micrometer screw should be rotated only in one direction to avoid backlash error.
5. The fringe width should be measured at a fairly large distance.
6. Convex lens of shorter focal length should be used ($f = 25$ cms. approx.)
8. Motion of eyepiece should be perpendicular to the lengths of the bench.

EXPERIMENT NO. ~~05~~ 10

Aim of the experiment: Determination of refractive index of the material of a prism using spectrometer and sodium light.

Apparatus required: source of light (sodium vapor lamp or mercury lamp), spectrometer, prism, spirit level.

Basic description of a Spectrometer: In some of the optics experiments, spectrometers are used. The spectrometer is an instrument for studying the optical spectra. Light coming from a source is usually dispersed into its various constituent wavelengths by a dispersive element (prism or grating) and then the resulting spectrum is studied. A schematic diagram of prism spectrometer is shown in Fig.1. It consists of a collimator, a telescope, a circular prism table and a graduated circular scale along with two verniers. The collimator holds an aperture at one end that limits the light coming from the source to a narrow rectangular slit. A lens at the other end focuses the image of the slit onto the face of the prism. The telescope magnifies the light dispersed by the prism (the dispersive element for your experiments) and focuses it onto the eyepiece. The angle between the collimator and telescope are read off by the circular scale. The detail description of each part of the spectrometer is given below.

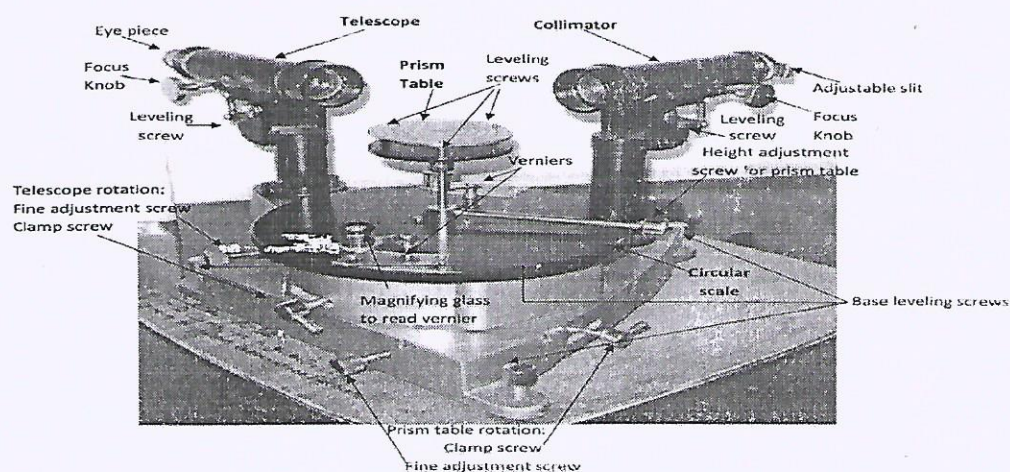


Fig. 1: Different parts of spectrometer

Physics Laboratory Manual

- a) **Collimator (C):** It consists of a horizontal tube with a converging achromatic lens at one end of the tube and a vertical slit of adjustable width at the other end. The slit can be moved in or out of the tube by a rack and pinion arrangement using the focus knob and its width can be adjusted by turning the screw attached to it. The collimator is rigidly fixed to the main part of the instrument and can be made exactly horizontal by adjusting the leveling screw provided below it. When properly focused, the slit lies in the focal plane of the lens. Thus the collimator provides a parallel beam of light.
- b) **Prism table (P):** It is a small circular table and capable of rotation about a vertical axis. It is provided with three leveling screws. On the surface of the prism table, a set of parallel equidistant lines parallel to the line joining two of the leveling screws, is ruled. Also, a series of concentric circles with the centre of the table as their common centre is ruled on the surface. A screw attached to the axis of the prism table fixes it with the two verniers and also keep it at a desired height. These two verniers rotate with the table over a circular scale graduated in fraction of a degree. The angle of rotation of the prism table can be recorded by these two verniers. A clamp and a fine adjustment screw are provided for the rotation of the prism table. It should be noted that a fine adjustment screw functions only after the corresponding fixing screw is tightened.
- c) **Telescope (T):** It is a small astronomical telescope with an achromatic doublet as the objective and the Ramsden type eye-piece. The eye-piece is fitted with cross-wires and slides in a tube which carries the cross-wires. The tube carrying the cross wires in turn, slides in another tube which carries the objective. The distance between the objective and the cross-wires can be adjusted by a rack and pinion arrangement using the focus knob. The Telescope can be made exactly horizontal by the leveling screws. It can be rotated about the vertical axis of the instrument and may be fixed at a given position by means of the clamp screw and slow motion can be imparted to the telescope by the fine adjustment screw.

Physics Laboratory Manual

d) **Circular Scale (C.S.):** It is graduated in degrees and coaxial with the axis of rotation of the prism table and the telescope. The circular scale is rigidly attached to the telescope and turned with it. A separated circular plate mounted coaxially with the circular scale carries two verniers, V1 and V2, 180° apart. When the prism table is clamped to the spindle of this circular plate, the prism table and the verniers turn together. The whole instrument is supported on a base provided with three leveling screws. One of these is situated below the collimator.

1. Measurement of angles with the help of spectrometer:

The spectrometer scales are angle measuring utilities for the positions of the telescope which can be rotated about the central axis of the instrument. The main circular scale is attached with the telescope so that when the telescope is rotated, the main circular scale also rotates with it. The angle, through which the telescope is rotated, can be measured by reading the positions of the verniers attached to the prism table and sliding over the main scale. In a spectrometer there are two sets of main circular scales (fitted with the telescope) and vernier scale (attached with the prism table). Both sets are diagonally (left hand and right hand sides) fixed in the instrument and measures angle for a particular telescope position with a difference of 180° . These scales can be used in a similar manner as a simple Vernier Caliper or traveling microscope is used. The vernier Caliper or traveling microscope is used to measure small distances (in centimeters and fractions) whereas spectrometer scales are used to measure small angular displacements (in degrees, minutes, and seconds) (1° is equal to 60 minutes, and 1 minute is equal to 60 seconds; ($1^\circ = 60'$ and $1' = 60''$))

2. Least Count of the Spectrometer Scale:

BIT, Physics Laboratory has two types of spectrometers in which

- (i) 60 divisions of vernier Scale are equal to 59 divisions of the Main Scale, and
- (ii) 30 divisions of vernier Scale are equal to 29 divisions of the Main scale.

Now, we will find out the least count in first case which 60 divisions of Vernier scale are equal to 59 divisions of the main scale. The method is as follows:

- a) Value of one division of circular main scale = $0.5^\circ = 30'$ (as $1^\circ = 60'$)

Physics Laboratory Manual

- b) Value of one division of sliding vernier scale = $(59/60) \times 0.5^\circ$
- c) Least count of spectrometer scale = Value of 1 div. of main scale – value of 1 div. of vernier = $0.5^\circ - [(59/60) \times 0.5]^\circ$
 $= [0.5 \times 1/60]^\circ = 0.5' = 30''$ (THIRTY SECONDS)
- d) Similarly, the least count of the spectrometer scale in second case in which 30 divisions of Vernier scale are equal to 29 divisions of the circular main scale can also be calculated. In this case the value of least count will be 1' or 60''.

3. Taking Readings on the two Spectrometer scales:

Following is an illustration for taking observation reading using the left hand side set of the circular main scale (attached with the telescope) and the corresponding vernier scale (sliding over the circular main scale and attached with the prism table). Assuming that we are using the spectrometer in which 60 divisions of vernier scale are equal to 59 divisions of the main scale.

The 0th division of the vernier scale precedes the circular main scale division whose value is 234° and 30'. Therefore, the main scale division reading is $234^\circ 30'$. Let 13th division of vernier scale coincides completely with a main scale division. Therefore, the vernier scale reading would be = 13x Least count of vernier scale

$$= 13 \times 30'' = 390'' = 6' 30''$$

$$\begin{aligned}\text{Total Spectrometer Scale Reading} &= \text{Circular Main scale Reading} + \text{Vernier Reading} \\ &= 234^\circ 36' 30''\end{aligned}$$

Reading of the right hand side scale can be similarly observed. The readings taken from left hand side and right hand side should ideally differ by 180°

Theory: A prism is a portion of refracting material bounded by three planes. A cross-section of prism by a horizontal plane is triangular in form. The prisms we use in the lab are equilateral prisms, but other configurations are also possible. The faces of a prism are called refracting faces. In the lab spectrometer experiment one face of the prism is etched or frosted so that light does not pass through it. The line at which two refracting faces meet is called a refracting edge. Since there are only two refracting faces, there is only one refracting edge.

Physics Laboratory Manual

A ray of light EF incident on one of the refracting faces gets refracted along the path FG through the prism and emerges along the path GH as shown in the Fig.2. The angle between the original direction of the incident ray and the emergent ray is called the angle of deviation D . For refraction through a prism,

$$i + e = A + D$$

where i & e are the angle of incidence and angle of emergence respectively, A being the angle of the prism and D is the angle of deviation.

Angle of deviation (D) depends upon the angle of incidence i . For a certain angle of incidence, the deviation will be minimum. It is denoted by D_m . The refractive index μ of the material of the prism is related to the angle of prism A and the angle of minimum deviation δ_m through the relation

$$\mu = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin\frac{A}{2}}$$

Where μ = refractive index of the material of the prism, A = angle of the prism, D_m = angle of minimum deviation. When the prism is at the minimum deviation position, the angle $i = e$ and the angles of refraction at both surfaces $r_1 = r_2$

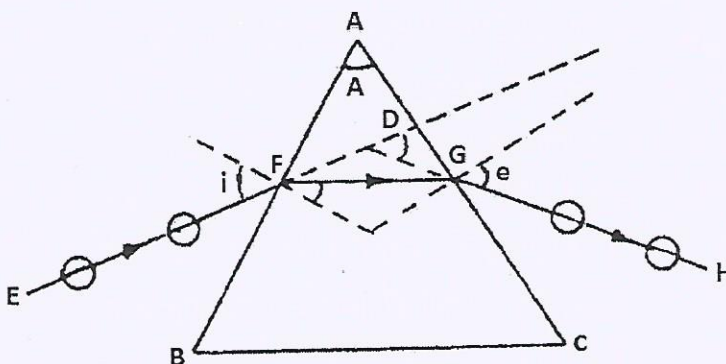


Fig. 2: Ray diagram of a ray of light through a prism

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$$\text{From } \text{ground} = i + e - A$$

$$r_1 + r_2 = A$$

$$D_m = 2i - A$$

or,

$$2i = A + D_m$$

$$i = A + D_m/2$$

$$\& r = A/2.$$

Procedure: Measurement of refractive index of material of the prism consists of two parts i. e.

A. Determination of the angle of prism A (For an equilateral prism we may take $A = 60^\circ$)

B. Determination of angle of minimum deviation D_m

Measurement of the angle of minimum deviation:

There are two procedures we may use to find D_m – a direct but approximate procedure, and a rigorous graphical procedure. In both cases we may use either monochromatic light from a sodium lamp, or polychromatic light from a mercury lamp. Note that the refractive index μ will differ for the different colours from the mercury lamp.

A) Direct procedure to find D_m for light from a sodium lamp, or from a mercury lamp

Initial procedures to practice and perform before main experiment:

Level the spectrometer base, prism table, telescope and collimator, using procedures explained elsewhere. Adjust the telescope and collimator for parallel rays as described elsewhere. Place the prism so that its center coincides with the center of the prism table (use the circles on the prism table to centre the prism). Adjust the prism so that light from the given source passes through the collimator and falls on one of the refracting faces of the prism and emerges out of the other refracting face after refraction. An image of the slit or the spectrum of light from the source is obtained, and may be viewed either directly by your eye or through the telescope.

Physics Laboratory Manual

Now remove the prism and bring the telescope in the line of the collimator. Observe the slit directly through the telescope and rotate the telescope so that the vertical crosswire coincides with the image of slit. Note the reading of any one of the two verniers – V_1 say – and continue to use only this one vernier scale throughout the experiment. Let the direct ray be D_o .

Place and centre the prism on the prism table as shown in the Fig. 3. If the telescope is placed as shown, keep the frosted surface to the left and the refracting edge to the right. We start with the procedure for monochromatic light from a sodium vapour lamp. Switch on the source and view the emergent light – image of the slit - with one unaided eye. Now adjust the prism for minimum deviation in the following way. Starting with the refracting face nearly normal to the light, rotate the prism table clockwise so that the refracting edge moves towards you, all the time keeping the image of the slit in view. You will easily find a position where the image or spectral line begins to recede in the opposite direction although you continue to rotate the table in the same direction. Once you find this position, you may rotate the prism both clockwise and anti- clockwise to find the same thing happening. This position where the spectral line begins to recede in opposite direction is the position of minimum deviation. Now while looking at the slit with the unaided eye, rotate the telescope so that it comes between the prism and the eye. Rotate the prism and if needed the telescope to find the position of minimum deviation. Note the reading of the vernier V_1 . Let it be R . The difference between the D_o and R is the minimum deviation D_m . Calculate refractive index from the equation (1) above.

If we use light from a mercury lamp please note that the light is polychromatic, and several coloured lines red, yellow, green, blue, will be seen. Focus on one colour at a time and repeat the above procedure for all colours (wavelengths).

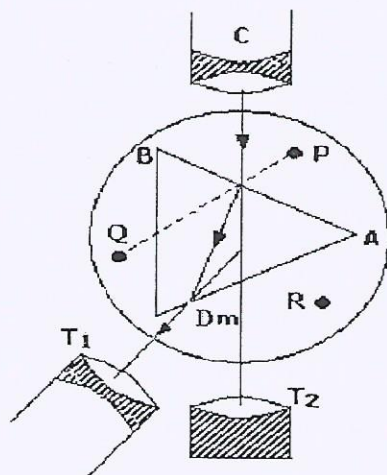


Fig. 3: Ray diagram for determination of angle of minimum deviation

B) Rigorous graphical procedure to find D_m for light from a sodium lamp

Initial procedures to practice and perform before main experiment:

Level the spectrometer base, prism table, telescope and collimator, using procedures explained elsewhere. Adjust the telescope and collimator for parallel rays as described elsewhere. Place the prism so that its center coincides with the center of the prism table (use the circles on the prism table to centre the prism). Adjust the prism so that light from the given source passes through the collimator and falls on one of the refracting faces of the prism and emerges out of the other refracting face after refraction. An image of the slit or the spectrum of light from the source is obtained, and may be viewed either directly by your eye or through the telescope.

Now remove the prism and bring the telescope in the line of the collimator. Observe the slit directly through the telescope and rotate the telescope so that the vertical crosswire coincides with the image of slit. Note the reading of any one of the two verniers – V_1 say – and continue to use only this one vernier scale throughout the experiment. Let the direct ray be D_o .

Place and centre the prism on the prism table. Facing the slit, keep the frosted surface of the prism to the left and the refracting edge to the right. Switch on the monochromatic light source - sodium

Physics Laboratory Manual

vapour lamp. Now take the readings for angle of incidence I on the prism and corresponding angle of deviation in the following way. Starting from the direct reading position D of telescope, rotate the telescope anti-clockwise by about 60° and then while looking through the telescope rotate the prism clockwise (so that the refracting edge moves towards you), until the image of the slit comes into view. Make accurate adjustments and take the reading P . Obviously $2I + (D \sim P) = 180$ and I can be calculated. Now without disturbing the prism, rotate the telescope clockwise and find the deviated ray. Take the reading R . Obviously $D \sim R = D$. Additional values of I and corresponding D may be taken by rotating the telescope in steps through $60^\circ + 2x$ from direct position (so that I increases by x , where $x = 10^\circ$ say), and finding the corresponding R and D .

While rotating the prism table note down the different angles from the vernier and read as angle of deviation and calculate the angle of incidence using the equation $i = A + D_m/2$.

1. Plot a graph (i vs D) taking angle of incidence i along the X-axis and angle of deviation D along the Y-axis. The nature of the graph is shown in Fig.4.
2. Draw a horizontal line as a tangent to the lowest point of the curve. Intersection of this horizontal line on Y-axis gives the angle of minimum deviation D_m (Fig.4).

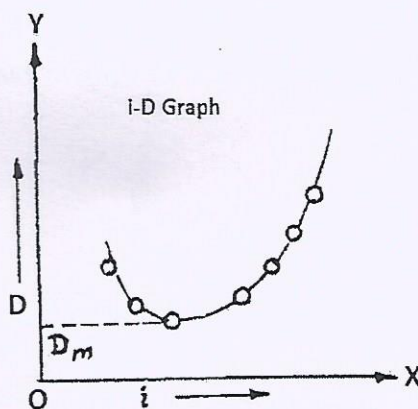


Fig. 4: Graph of angle of deviation vs angle of incidence

Physics Laboratory Manual

Observations:

The angle of given prism is 60°

Table 1: Calculation of the angle of minimum deviation (D_m):

S.No.	Colour	Vernier	Telescope Reading for minimum deviation			Telescope Reading for direct image			Difference $D_m = a-b$	Mean Value of D_m
			MSR	VSR	TR(a)	MSR	VSR	TR(b)		
1.	Violet	V ₁								
		V ₂								
2.	Red	V ₁								
		V ₂								
3.	Yellow	V ₁								
		V ₂								

MSR = Main Scale Reading, VSR = Vernier Scale Reading, TR = MSR+VSR = Total Reading

Calculations: Putting the mean value of A from Table 1 and the angle of minimum deviation D_m from the graph, the refractive index of the material of the prism can be found out.

Result: Refractive Index for the material of the prism:

S.No.	Color	Calculated Refractive Index	Standard Refractive Index	%Error

Physics Laboratory Manual

Precautions and Sources of Error:

1. The telescope and collimator should be individually set for parallel rays.
2. Slit should be as narrow as possible.
3. While taking observations, the telescope and prism table should be clamped with the help of clamping screws.
4. Both verniers should be read.
5. The prism should be properly placed on the prism table for the measurement of angle of the prism as well as for the angle of minimum deviation.